



Investment Society
Imperial College London



Lecture 2

Securities Education Certificate

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Slide deck inspired by Dr. Euan Doidge and Prof. Anthony Barrett (Department of Chemistry, Imperial College London)

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Background

Some context to introduce you to a concept

Fact

You will need to recall this information in the exam

Model Example

A fully detailed breakdown of an example (question)

Try it yourself

It's your turn to give this problem a go and test your knowledge!

Equation sheet

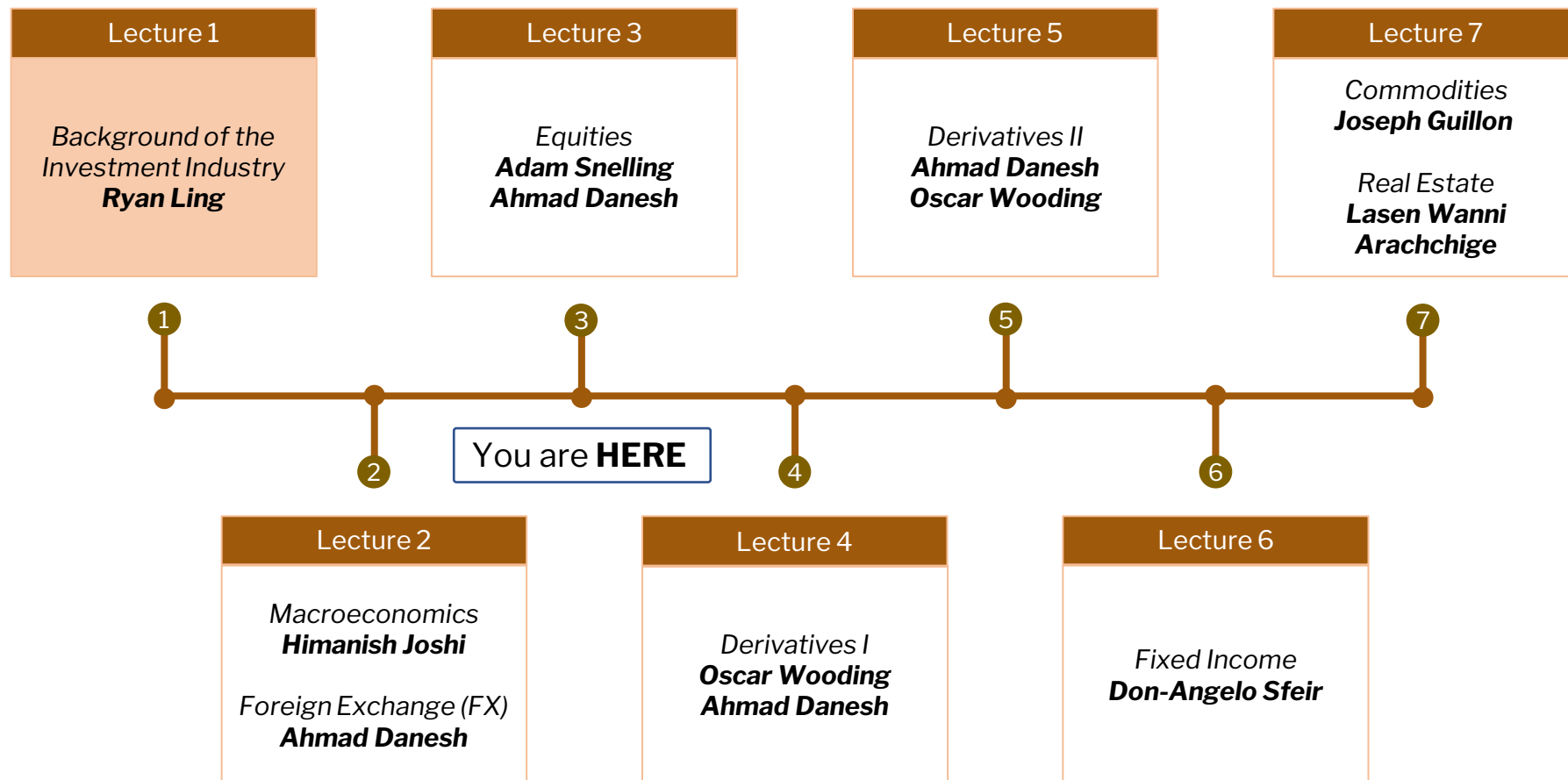
Any equations that you may need for the exam are provided

Taking it further

(non-examinable) stuff to show you where this concept can be applied

The slides are colour-coded to aid your understanding of the course and ensure that you are familiar with the demands of the content provided in each slide.

Lecture Timeline



Let's begin...



Himanish Joshi

Lecturer

*Sector Head, QT Capital
Financials*

Lecture 2

Macroeconomics

Macroeconomics

Economics is divided into 2 sections:

Microeconomics

“Bottom-Up” perspective.



Considers the economy at an individual and business level, i.e. singular products or services.

Key Topics: Types of Economy, Demand & Supply, Market Equilibrium

Macroeconomics

“Top-Down” perspective.



Considers the economy at a country and international level e.g. governments and central banks' policies.

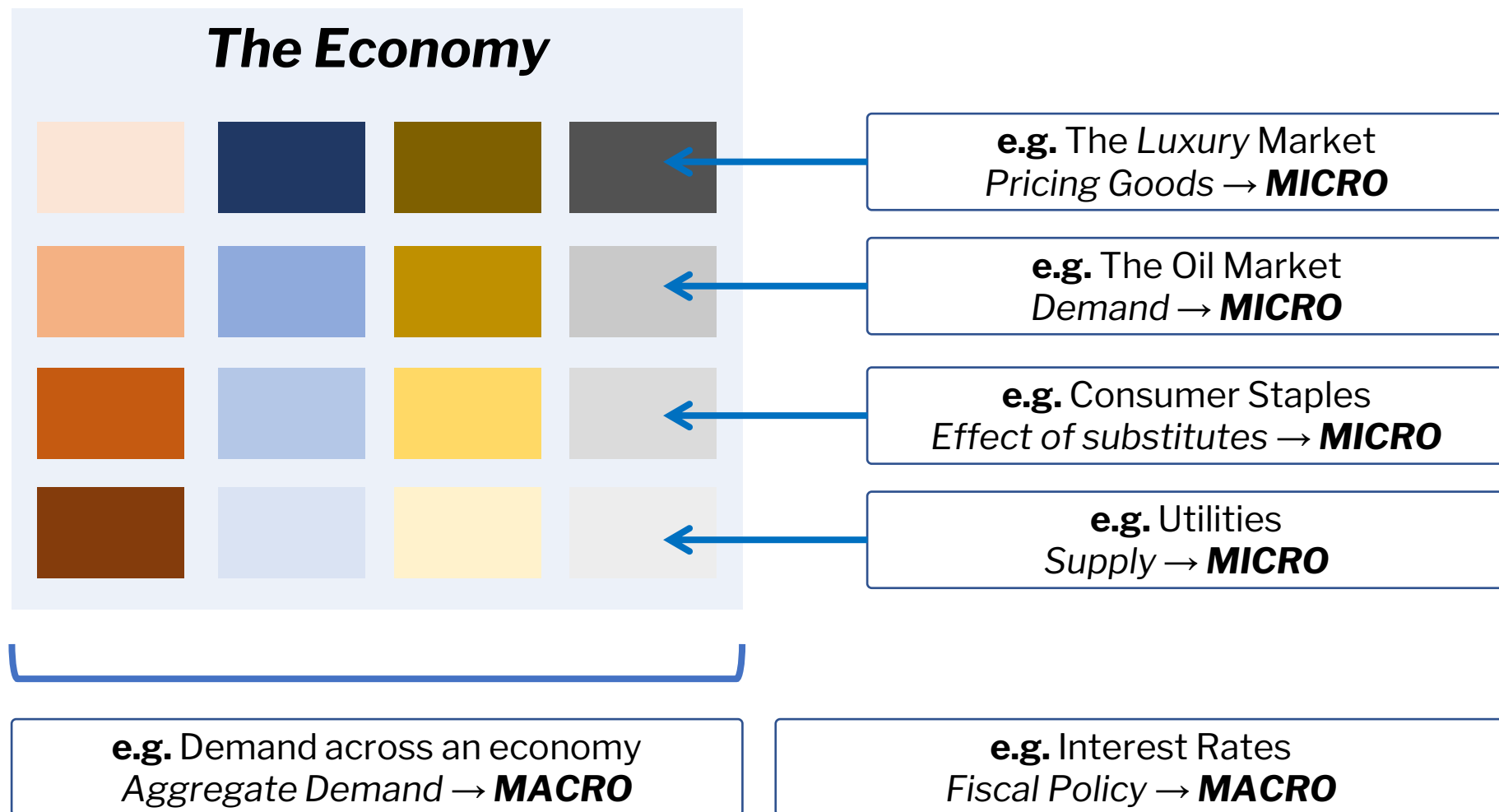
Key Topics: Aggregate Demand & Supply, Inflation, Fiscal and Monetary Policy

👉 To understand Macro, we first need to look at some Micro concepts.

Your notes

Macroeconomics

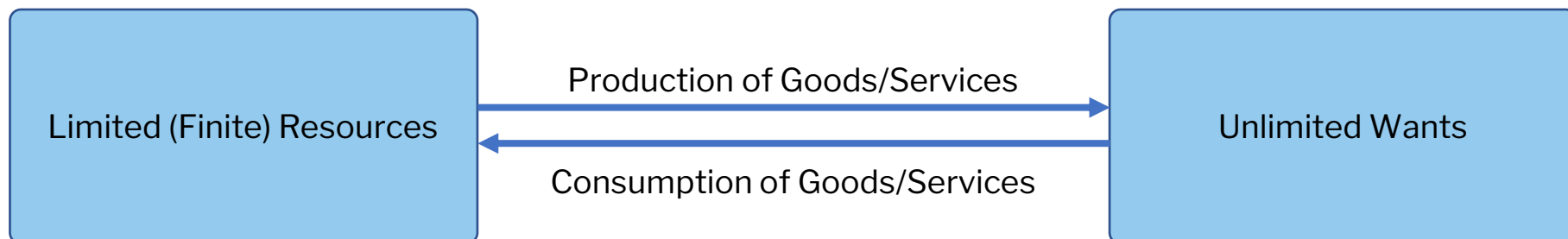
Here's a way to think about the difference between macro- and microeconomics.



Fundamental Concepts

The Fundamental Economic Problem is that of Scarcity.

👉 **Scarcity** is the idea that we have unlimited wants but limited resources to fulfill those wants.



👉 This means decisions need be made:

What do we need to produce?

How much do we need to produce?

👉 Who and how these decisions are made depend on the type of economy.

Your notes

Demand

Let's tackle the first big idea of economics... what is demand?

👉 **Demand** is how much consumers are *willing* and able to buy at different prices **AT A GIVEN TIME**.

👉 **Price** is what a buyer pays for one unit of a specific good/service.

👉 **Quantity Demanded** (Q_D) is the number of units purchased at this price.

👉 Inverse relationship between price and quantity demanded → '**Law of Demand**'. 3 reasons*:

Diminishing Marginal
Utility*

*Your utility of each additional unit consumed **decreases**.*

Income Effect

Consumers have fixed budgets.

Substitution Effect

If something is too expensive you will find an alternative.

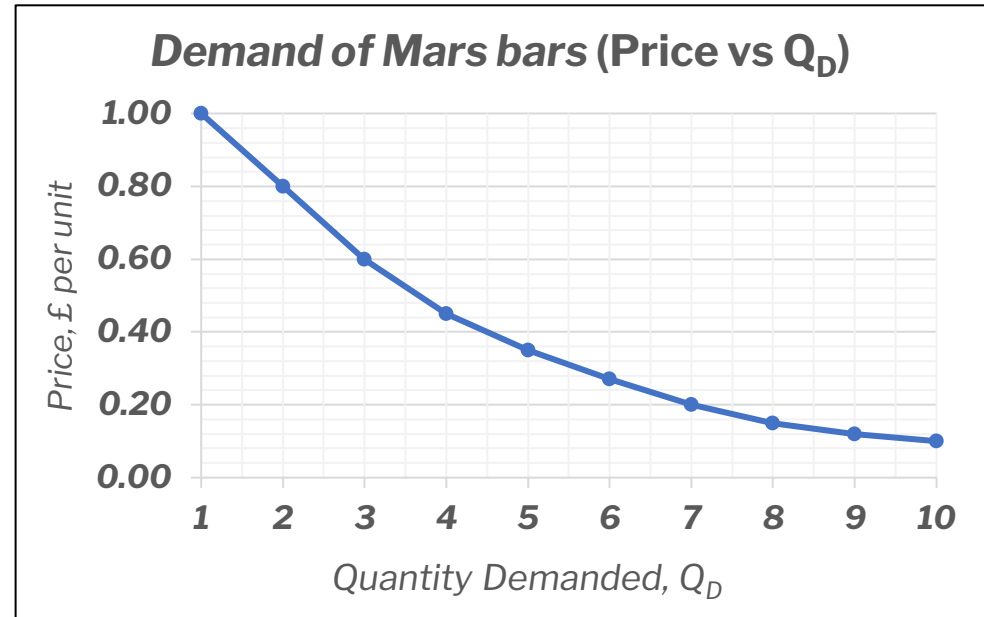
**For a more detailed explanation of the 3 key reasons please see the notes.*

Demand Curves

*Law of demand lets us create demand schedules, which give us **demand curves**!*

Consider a market for Mars bars. We plot the quantity on the x-axis and price on the y-axis.

Price (£ per unit)	Quantity Demanded, Q_D (no. of units)
1.00	1
0.80	2
0.60	3
0.45	4
0.35	5
0.27	6
0.20	7
0.15	8
0.12	9
0.10	10

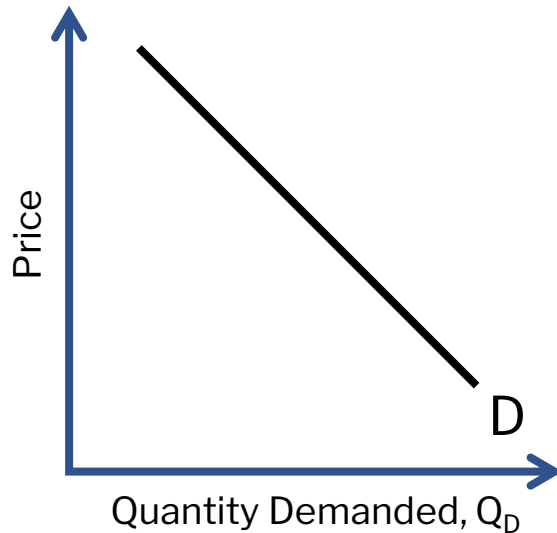


Your notes

Demand Curves

So, that was the demand for Mars bars. What about other goods/services?

Simplified **Demand** curve
(applies to any good/service)
due to the *Law of Demand*



☞ Demand curves for different goods/services will have different shapes/curvatures.

☞ BUT the law of demand applies in every case!

☞ All demand curves go from top left to bottom right.

☞ Demand curves can be simplified to straight lines.

☞ SO we can approximate the demand curve D for any good/service as a downward sloping line.

Your notes

Demand Curve Movements and Shifts

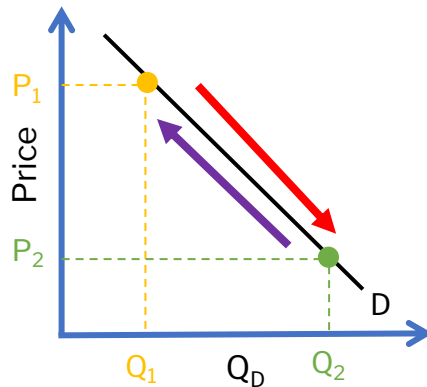
You can model macroeconomic events by considering changes in the demand curve!

👉 A **movement along the demand curve** can only be caused by a change in the price.

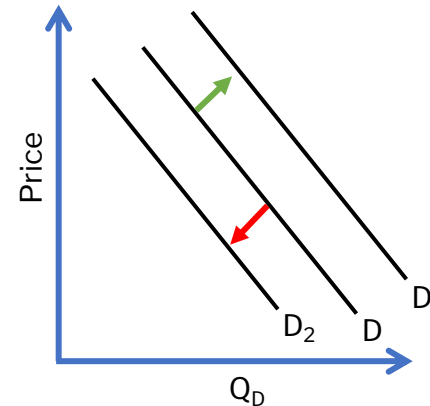
👉 This results in us operating at a different point along the same curve **D**.

👉 A **shift of the demand curve** occurs when *any factor other than price* changes demand.

👉 New curves D_1 and D_2 drawn parallel to **D** as Q_D has changed for every price.



Purple: a **contraction** in demand
Red: an **extension** in demand



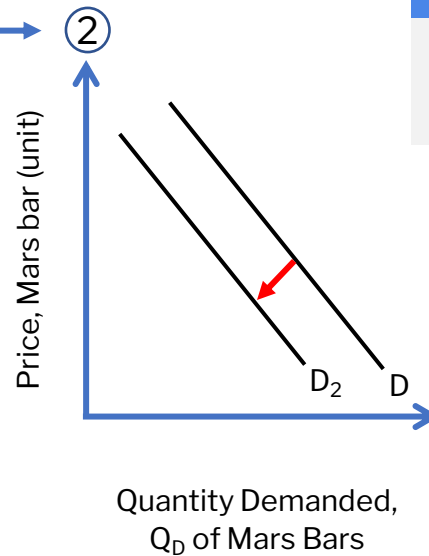
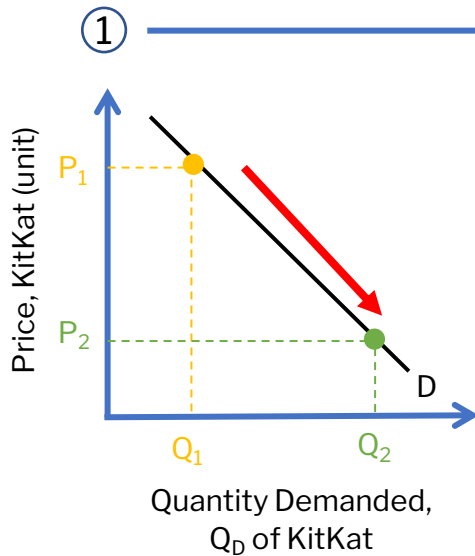
Green arrow: an **increase** in demand (shift out)
Red arrow: a **decrease** in demand (shift in)

Your notes

Changing Demand

Think about it... what would cause a shift in demand?

If e.g. the price of KitKats go down, the law of demand tells us that Q_D of KitKat **increases**.



So, consumers will have less money to spend on Mars bars → Q_D of Mars bars will **decrease**.

☞ A change in the price of substitute goods caused a change in demand.

☞ Other factors can also affect the quantity demanded (Q_D):

Changes in Income

Changes in Population

Changes in Trends

Supply

Demand is seen through the consumer. Supply is seen through the producer.

☞ **Supply** is how much suppliers are *willing* and able to produce at different prices **AT A GIVEN TIME**.

☞ Firms are driven by **profit** → If the price of the good/service increases, profit increases.

☞ A positive between price of a good/service and quantity supplied → ‘Law of Supply’.

If the market price of Mars Bars go up, Mars will manufacture more Mars bars to take more profit.

☞ **N.B.** When there is a profit motive, we assume we are operating in a ***free-market economy**.

Your notes

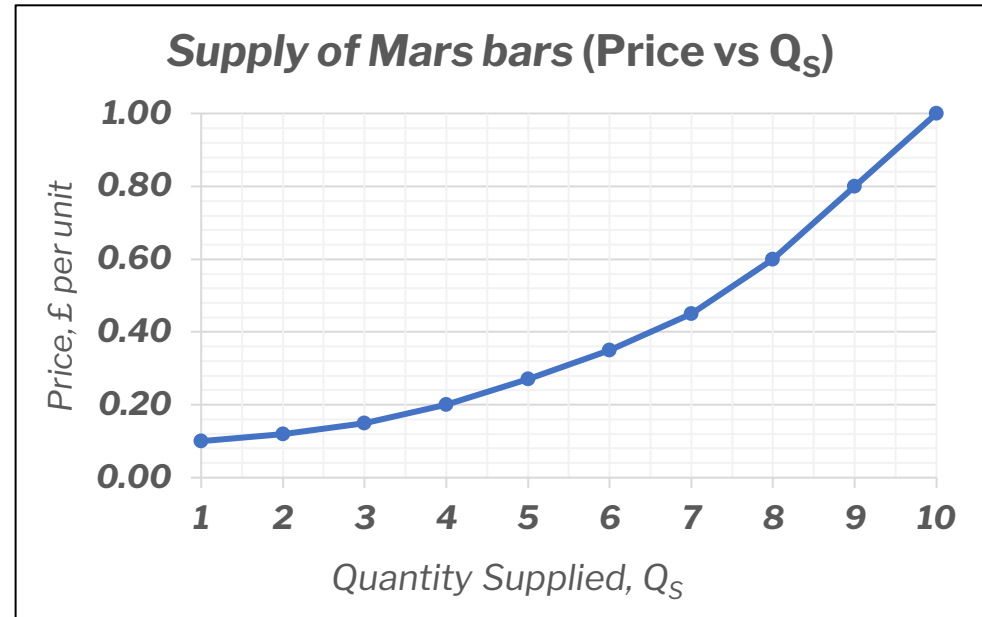
**Free-market economies are discussed further in the notes.*

Supply Curves

Just like demand, supply schedules can let us plot supply curves!

Again, consider a market for Mars bars. We plot the quantity on the x-axis and price on the y-axis.

Price (£ per unit)	Quantity Demanded, Q_D (no. of units)
0.10	1
0.12	2
0.15	3
0.20	4
0.27	5
0.35	6
0.45	7
0.60	8
0.80	9
1.00	10

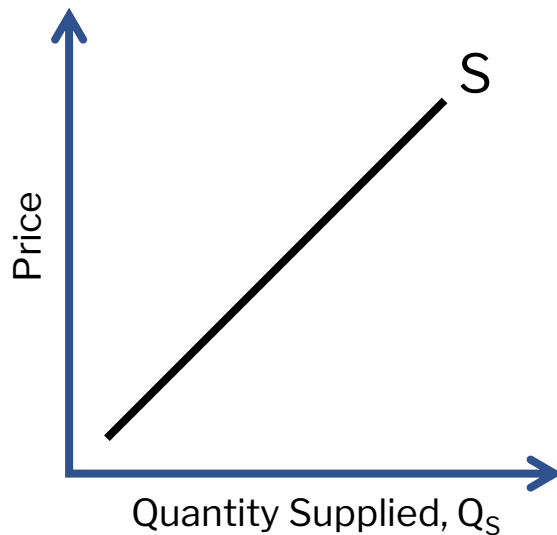


Your notes

Supply Curves

Once again, we can make generalisations about the supply curves for any good.

Simplified **Supply** curve
(applies to any good/service)
due to the *Law of Supply*



☞ Supply curves for different goods/services will have different shapes/curvatures.

☞ BUT the law of supply applies in every case!

☞ All supply curves go from bottom left to top right.

☞ Supply curves can be simplified to straight lines.

☞ So, we can approximate the supply curve S for any good/service as an upward sloping line.

Your notes

Supply Curve Movements and Shifts

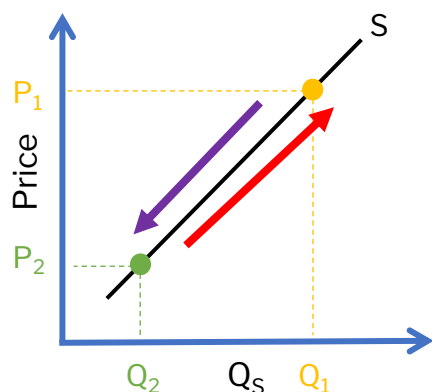
Supply curves can also change to reflect changing conditions via movements and shifts.

👉 A **movement along the supply curve** can only be caused by a change in the price.

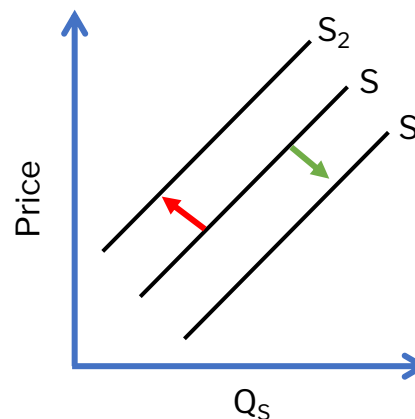
👉 This results in us operating at a different point along the same curve **S**.

👉 A **shift of the supply curve** occurs when any *factor other than price* changes supply.

👉 New curves S_1 and S_2 drawn parallel to **S** as Q_s has changed for every price.



Purple: a **contraction** in supply
Red: an **extension** in supply



Green arrow: an **increase** in supply (shift out)
Red arrow: a **decrease** in supply (shift in)

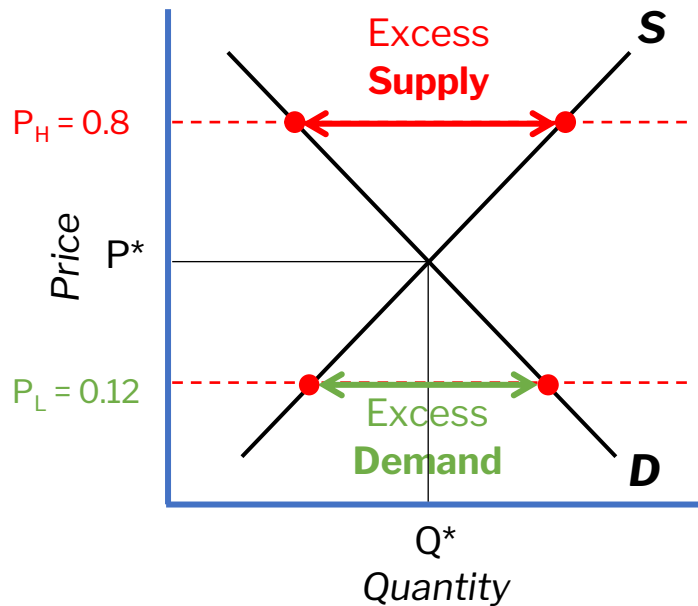
Your notes

Market Equilibrium

Plotting the supply and demand curves together shows the market for a good/service.

👉 Plotting both demand and supply curves gives us an intersection i.e. **market equilibrium**.

👉 Here, the quantity demanded by consumers is balanced by the quantity that firms wish to supply.



👉 At price P_H supply > demand. To profit of the excess supply, the producer has to lower prices.

👉 At price P_L demand > supply. To remove the excess demand, prices have to be raised.

👉 At price P^* , demand = supply → The market 'clears'. This is **market equilibrium**.

👉 **So**, markets always clear at equilibrium as this is the most efficient.

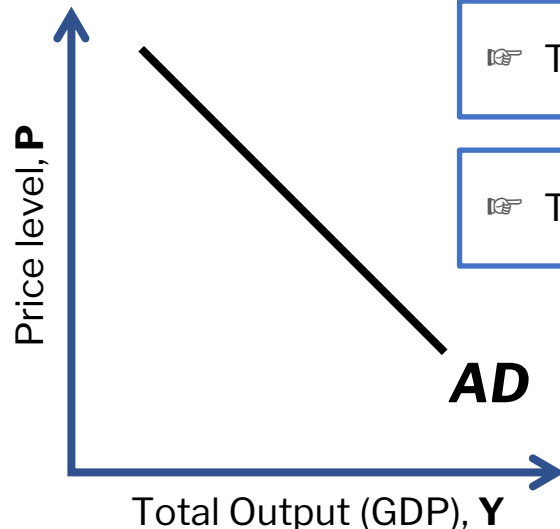
Aggregate Demand (AD)

We can now start looking at macroeconomics by first considering Aggregate Demand (AD).

👉 **Aggregate Demand (AD)** concerns the total spending of consumers in an economy at a given time.

👉 Aggregate Demand looks at the demand for all of the goods and services in an economy.

👉 The Law of Demand applies! → AD is shaped **exactly** like a normal demand curve.



👉 The x-axis represents **total output (GDP)** or **employment** with symbol Y.

👉 The y-axis represents **price levels** (and, loosely speaking: “inflation”).

👉 **Price levels** show the average price of goods or services.

Components of AD

What components make up Aggregate Demand?

$$AD = C + I + G + (X - M)$$

👉 **Consumption (C):** Expenditure on goods/services by households (ca. ~70% of AD).

👉 **Investment (I):** Expenditure by firms to add to their capital inventory (machinery, factories etc.)

👉 **Government Spending (G):** e.g. welfare spending and public-sector spending.

👉 **Net Trade (X-M):** Some goods paid for are imports (**M**) and some goods are exports (**X**).

👉 Any change in the factors above lead to a change in AD → a **shift** of the AD curve.

Aggregate Supply (AS)

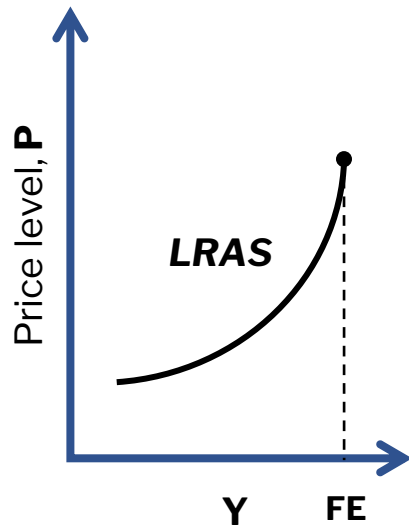
There are 2 forms of AS, SRAS (short run) and LRAS (long run). Here, we will consider LRAS*.

👉 **Aggregate Supply (AS)** concerns the total value of goods supplied by producers at a given time.

👉 Aggregate Supply looks at the supply of all of the goods and services in an economy.

👉 The Law of Supply applies → LRAS is an upward sloping curve*.

LRAS is **NOT**
a straight line!



👉 The x-axis represents **total output (GDP)** or **employment** with symbol Y.

👉 The y-axis represents **price levels** and (loosely speaking) “inflation”.

👉 FE denotes (full employment) – all resources in the economy are used.

**For a more detailed explanation of the difference between SRAS, LRAS and their respective curve shapes, see the notes.*

Inflation – What is it?

Undoubtedly, things seem to be costing more these days...

👉 **Inflation** is the *sustained* rise in general price levels over time in an economy.

👉 **HIGH** inflation can have a number of different effects:

Increases
cost of living

Decreases
buying power

Disrupts
business planning

Discourages
investment by firms

👉 Very low or negative inflation (**deflation**) can also be bad as it delays consumer spending.

👉 Overall, low and stable inflation is a positive for the economy.

2%
UK target inflation rate

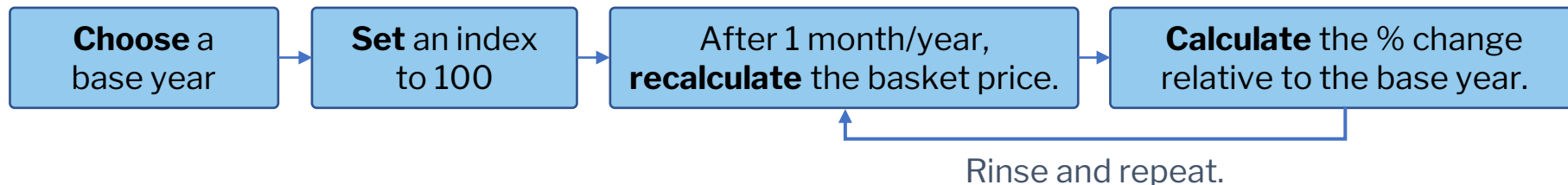
Your notes

CPI Inflation – How do we measure it?

What does an inflation rate of 2% mean?

👉 The % change in the **Consumer Price Index (CPI)** helps measure inflation.

👉 CPI measures price levels in the economy by considering a weighted basket of goods/services.



CPI v Core CPI

Countries report inflation figures with the CPI inflation rate and a Core CPI Inflation rate.

👉 Normal **CPI** reading (often termed “**headline inflation**”) covers most domestic consumption.

👉 It often has volatile readings due to factors not caused by economic conditions.

Food and Energy can be extremely volatile due to:

Natural Disasters

Political Turmoil

Seasonality

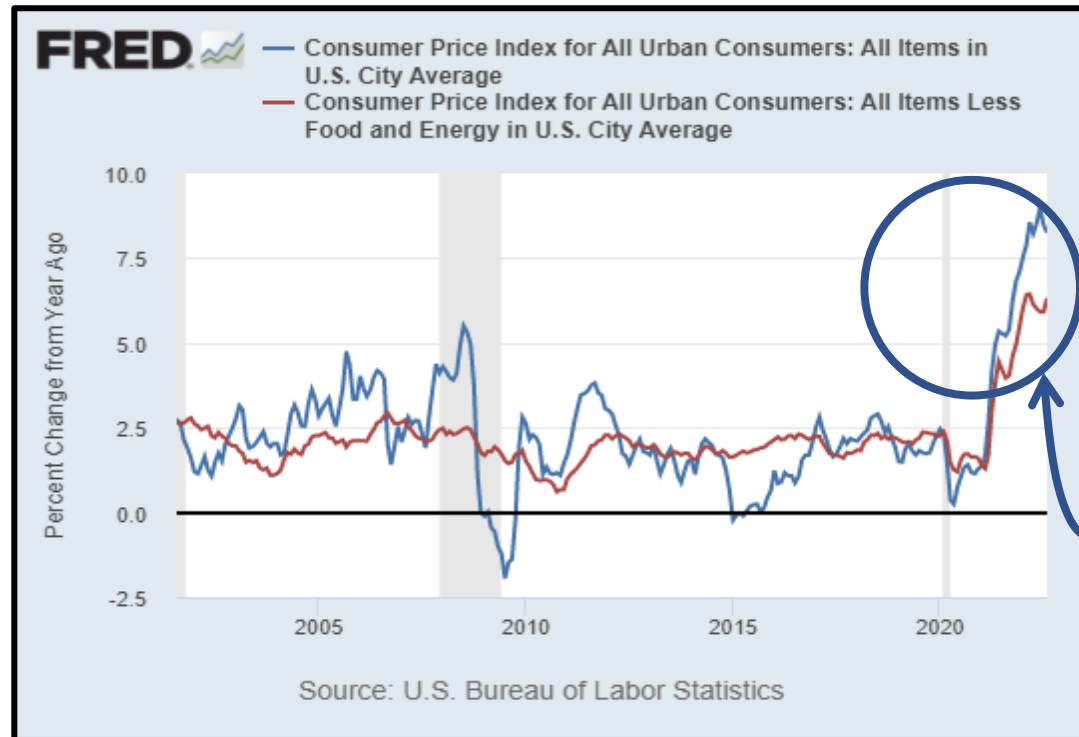
👉 **Core CPI** removes the food and energy components.

👉 Core CPI gives a more accurate reflection of economic conditions.

Your notes

US CPI v US Core CPI

A graph showing CPI and core CPI in the US over the past 20 years.



☞ Core CPI is more stable than CPI.

☞ Policy-makers often use Core-CPI as a *barometer* for price-levels in the economy.

NOTE Rapid rise in CPI and Core CPI (2020 onwards)

Image Source (October 10, 2022)

FRED Federal Reserve Bank of St Louis

Data Source US Bureau of Labor Statistics

Your notes

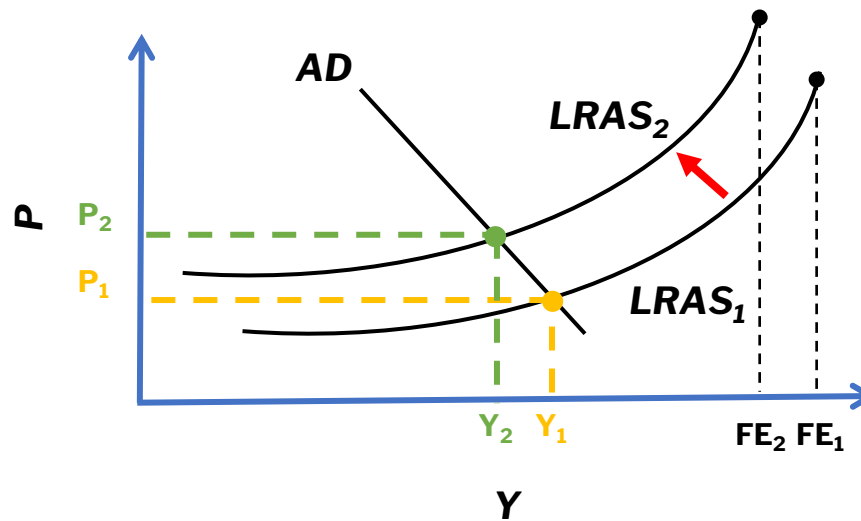
Supply Side Inflation

One type of inflation is called supply-side inflation (A.K.A. “*Cost-Push*” inflation).

👉 **Supply-side inflation** is caused by rising costs of production which leads to decreasing LRAS.

The war in Ukraine caused rising raw material costs → fewer goods/services → decrease in LRAS.

👉 Supply-side problems are currently present in many economies across the world.



Your notes

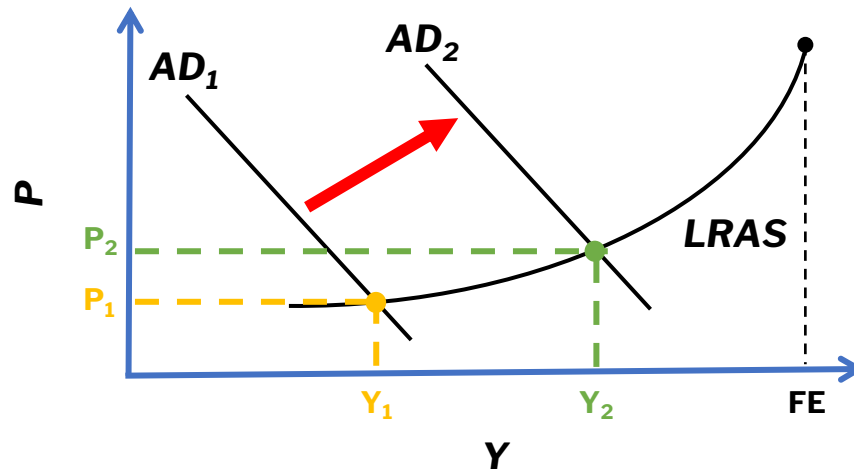
Demand Pull Inflation

The second type of inflation is called “Demand-Pull” inflation.

👉 **Demand-pull inflation** is where AD has increased, raising prices and output in the economy.

After COVID lockdowns, consumer spending increased in e.g. restaurants, travel and entertainment.

👉 **Consumption** increasing led to an increase in AD. This “pulled” prices and employment higher.



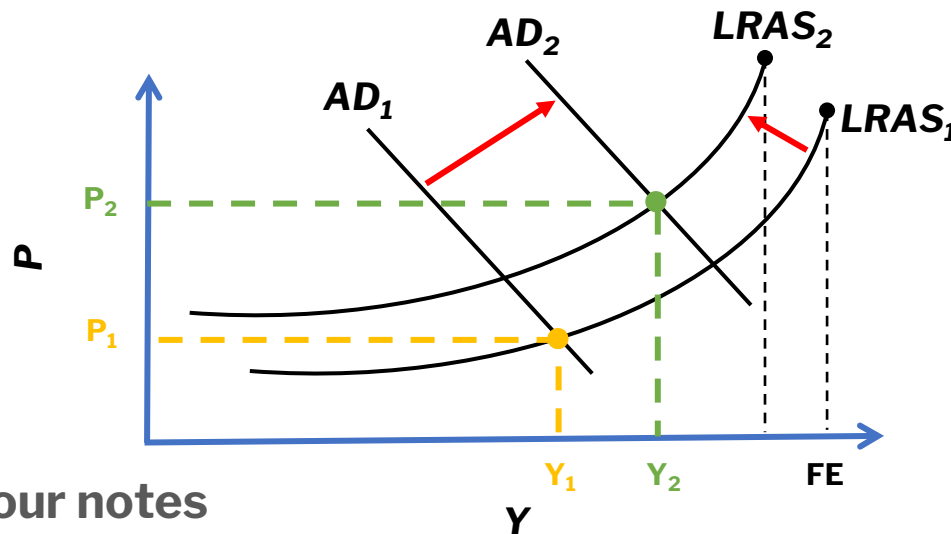
Your notes

Current Macroeconomic Environment

Aren't we currently experiencing *both* supply-side and demand-pull inflation?

👉 **Yes.** Most economies are facing inflation from both demand and supply sides.

👉 The diagram below is a good representation of current macro environments.



👉 The combination of **demand** and **supply-side** inflation has led to double-digit inflation in many countries.

How do governments tackle this inflation?

Your notes

Fiscal Policy

There are 2 forms of government policy to tackle inflation.*

Fiscal Policy

Monetary Policy

👉 **Fiscal Policy** involves taxes and government spending.

👉 Reducing income taxes or VAT can increase consumption (**C**) in the economy → higher AD**

👉 Reducing corporation taxes can lead to higher investment spending by firms (**I**) → higher AD

👉 Increasing Government Spending (**G**) → higher AD.

Increased welfare spending

Higher **spending** on public services

*There are many other policy objectives as well e.g.: economic growth, employment, public services, quality of life...

**The reverse also applies in all cases. So, increasing taxes lead to lower disposable incomes and less consumption.

Monetary Policy – Interest Rates

The key tools in Monetary Policy are interest rates and quantitative easing.

☞ **Interest Rates** determine the **cost of borrowing** in the economy for firms and consumers.

☞ They have a wide ranging effect on the economy through many different channels.

‘Propagation’ mechanisms (below) are a great way to map monetary policy!

☞ High Interest Rates → High costs of borrowing → Less spending → Saving is more attractive.

☞ Low Interest Rates → Low costs of borrowing → More spending → Saving is less attractive.

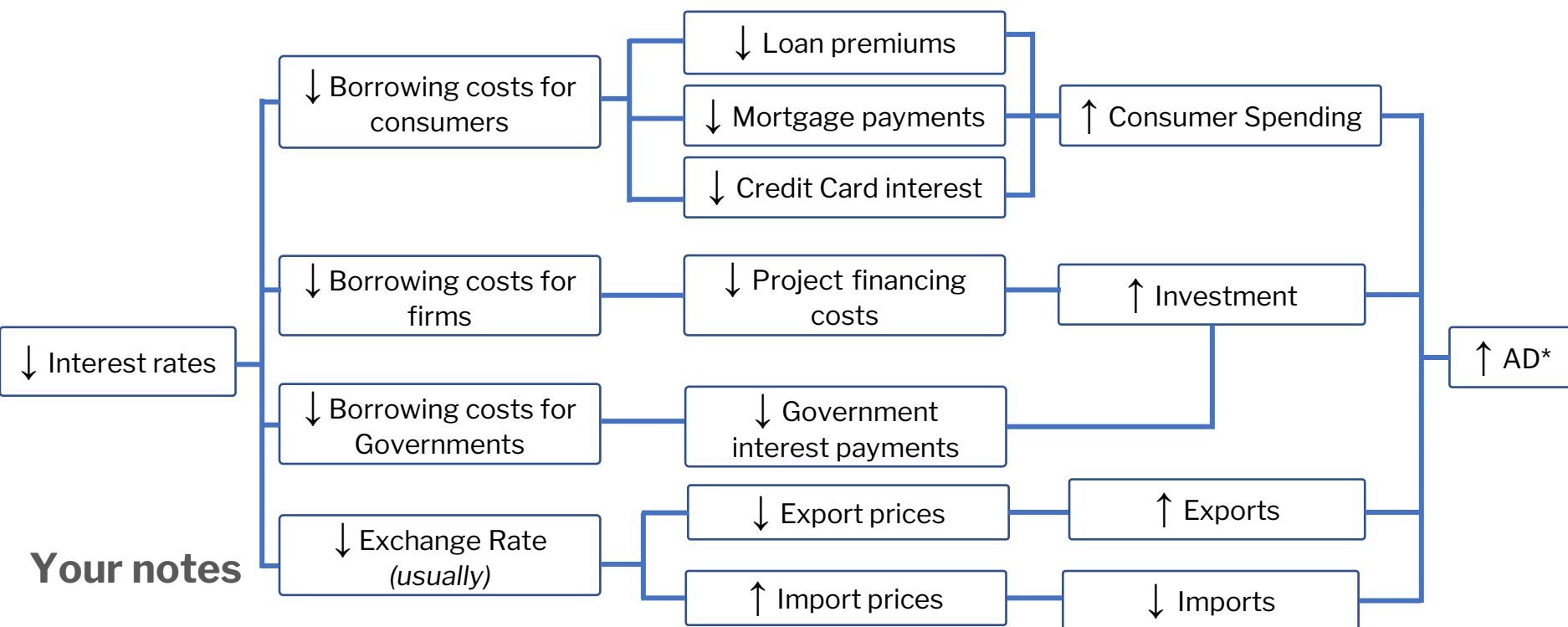
Your notes

**The reverse also applies in all cases. So, increasing taxes lead to lower disposable incomes and less consumption.*

Monetary Policy – Interest Rates

Let's consider a central bank e.g. BoE reducing interest rates to increase AD.

👉 Reducing interest rates is typically done during **economic downturns**. e.g. COVID lockdown, 2020.



Your notes

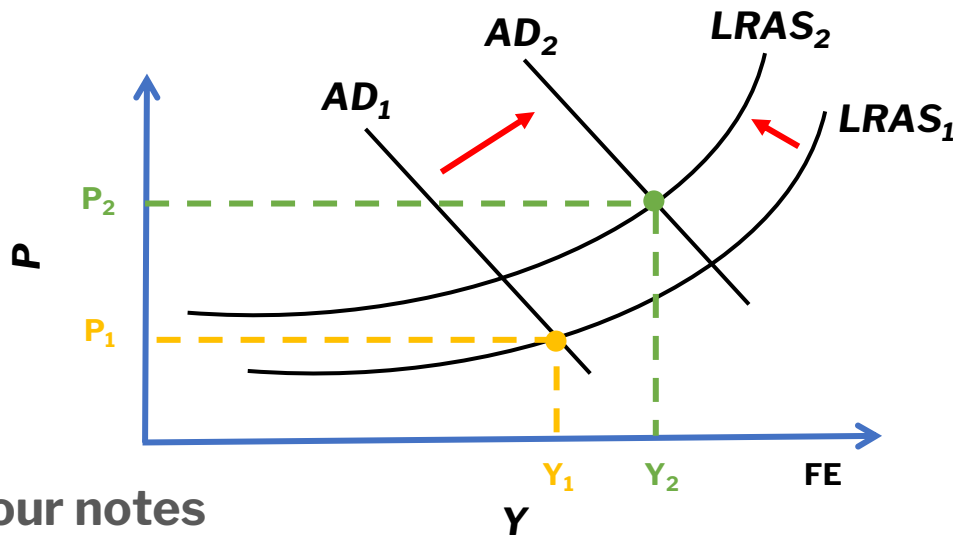
**There are also effects on the supply side (LRAS) – see notes.*

Explaining the Recent Financial Turmoil

We can now put theory into practice to explain what recently spooked Markets.

Consider what's happening right now from a macroeconomic perspective.

👉 **Inflationary pressures** due to combined demand and supply side factors.



What should the government trying to be doing to control inflation?

What fiscal policy was proposed in September budget?

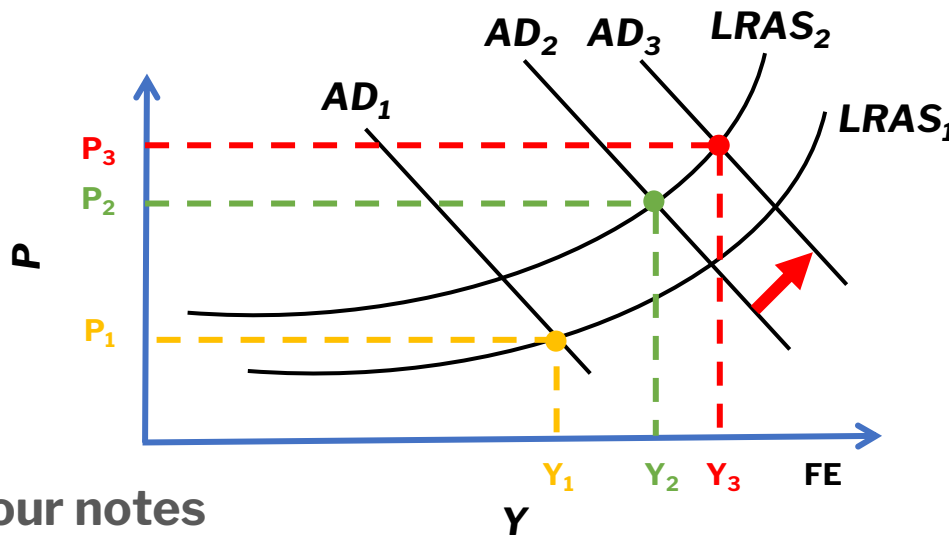
Your notes

Explaining the Recent Financial Turmoil

We can now put theory into practice to explain what recently spooked Markets.

👉 The mini-budget in September included **decreases in taxes** and **increased government spending**.

👉 This results in increasing AD (outward shift to AD_3) **making inflation worse** (P_2 to P_3)!



👉 High inflation increases costs of living and decreases purchasing power.

👉 The budget policy included helping low-income homes with high bills.

Here, only impacts on **inflation** are shown.

Your notes

Explaining the Recent Financial Turmoil

But, what about the Monetary Policy action by the BoE?

👉 The BoE has been aggressively increasing rates to combat inflation.

👉 The markets believed the mini-budget was terrible: Higher inflation → higher BoE interest rates.

👉 The market **lost confidence** in the UK policy, leading to huge sell-offs in the GBP and UK Gilts.

👉 High interest rates are damaging for economies in the long term.

👉 Decreased consumption (**C**) + decreased investment (**I**) → Lower AD → slower economic growth.

Your notes

*Let's resume in **5** minutes...*



Ahmad Danesh
Lead Lecturer
Founder, QT Capital

Lecture 2

Foreign Exchange

Foreign Exchange

Macroeconomics can apply to a wide range of markets.
Let's start with **foreign exchange**.

Donald Trung Quoc Don (Chữ Hán: 徵國單) - Wikimedia Commons

A Bureau de Change in Amsterdam, Netherlands.



Currency Exchange		
	WE BUY	WE SELL
USD/EUR 	1,2195	1,1494
USD 100 	82,00	87,00
GBP 100 	112,00	119,50
CHF 100 	88,00	94,50

NO COMMISSION AT ALL!

Foreign exchange **markets** are places where you can exchange one currency for another.

For retail investors, one example is at a *Bureau de Change* (pictured, left – in Amsterdam.).

Here, you can buy \$1.1494 for €1 in order to buy a *foreign* currency.

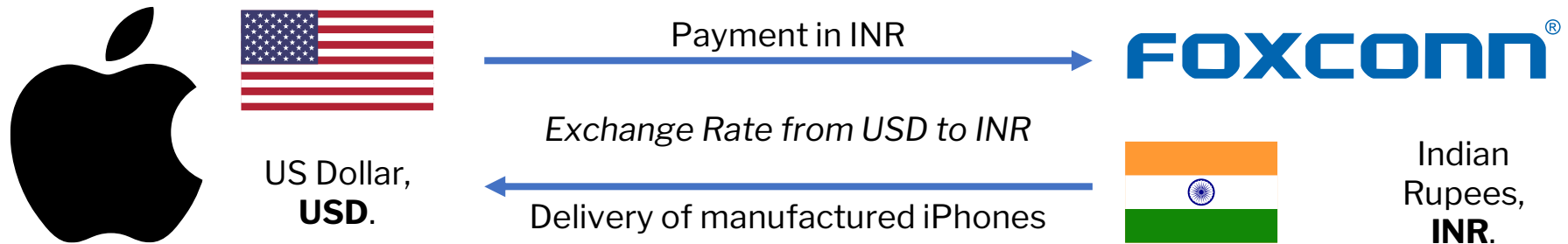
Alternatively, you can sell \$1.2195 to get €1 in order to buy a *local* currency.

Your notes

Globalisation

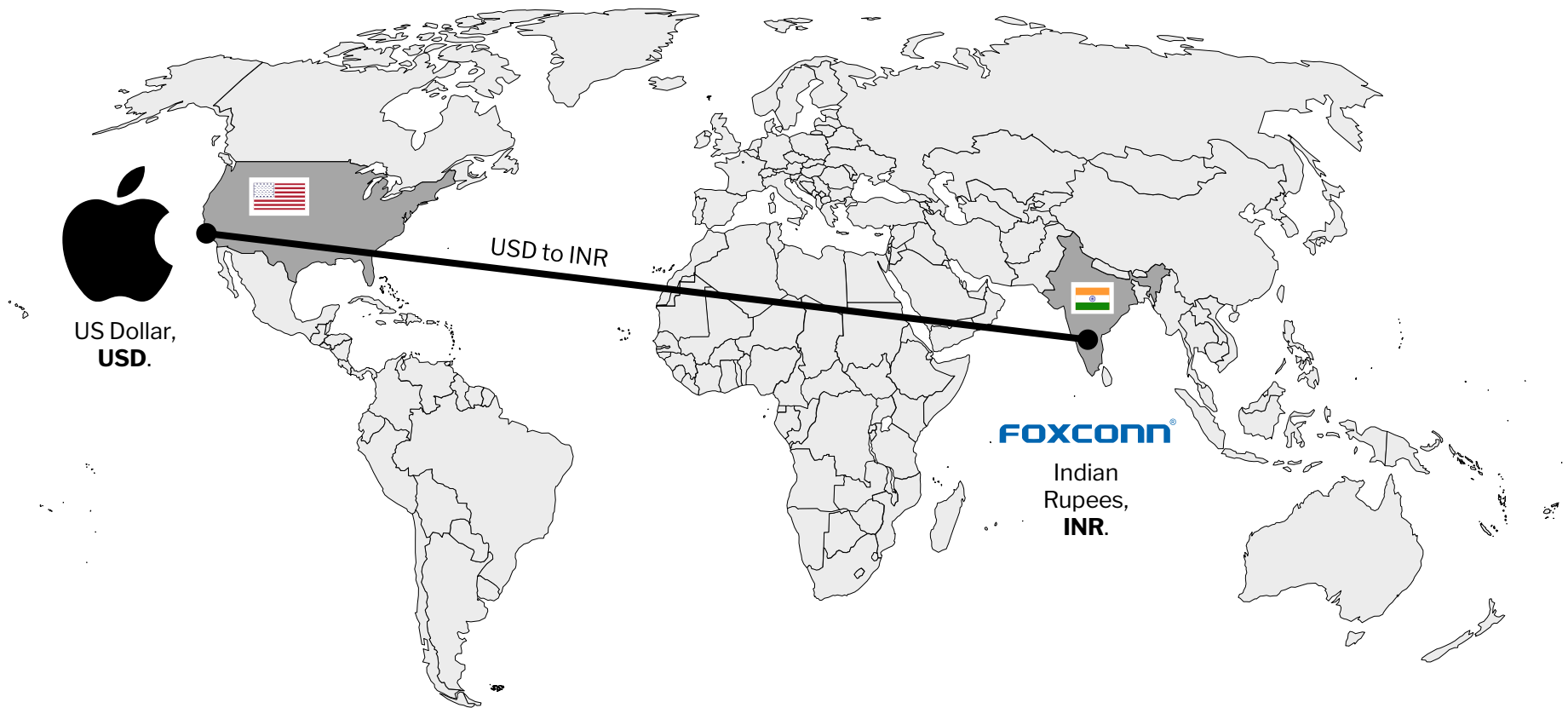
Why are foreign exchanges relevant in markets?

- ☞ A company based in one country may make its products in another country → globalisation.
- ☞ Production costs must be paid in the local currency of the country where the factory is located.
- ☞ Apple (located in the USA) pay Foxconn factories in India for iPhone production in **Rupees, INR**.



Your notes

Globalisation

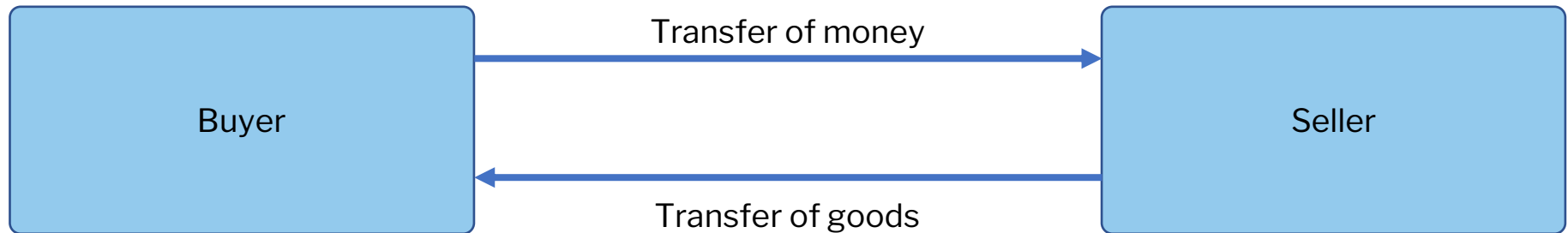


Your notes

Exchange Rates

But how are prices of currencies quoted? This is where exchange rates help.

👉 Remember that any trade can be simplified to two parties: a **buyer** and a **seller**.



👉 This idea is broadly expanded to when currencies are traded.

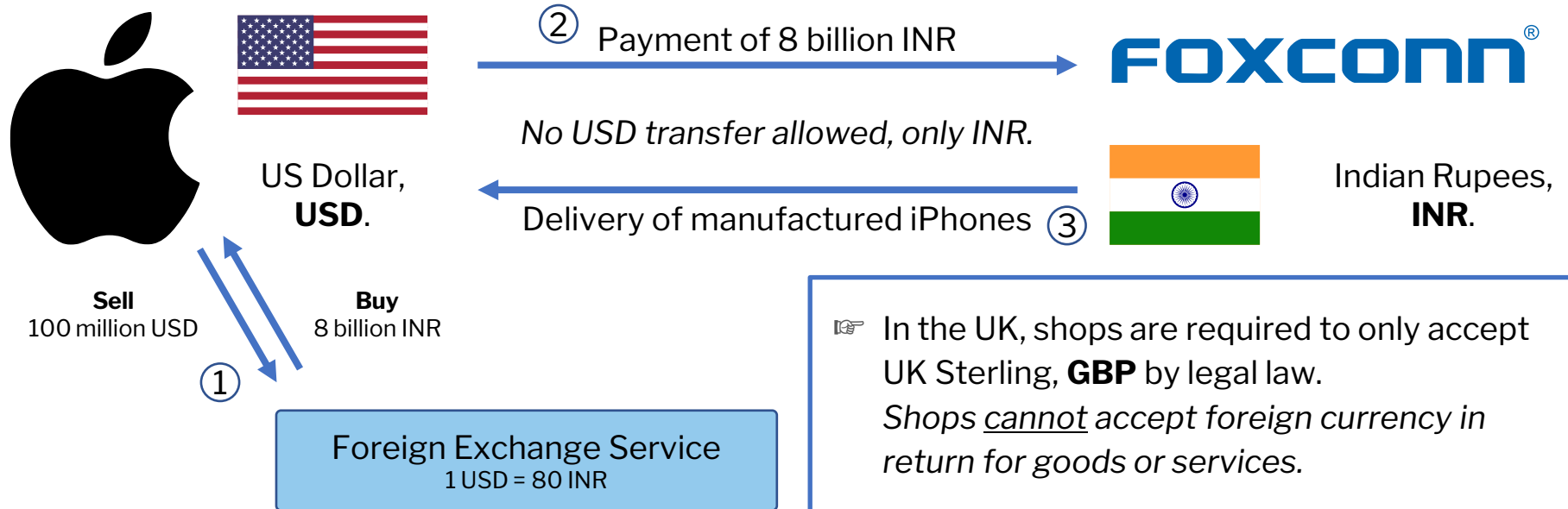
👉 One currency is being *bought* while another is being *sold*. We call this a **currency pair**.

Your notes

Exchange Rates

How does this compare in practice?

👉 If Apple is billed 8 billion INR, and 1 USD = 80 INR, Apple must convert 100 million USD to INR.



👉 In the UK, shops are required to only accept UK Sterling, **GBP** by legal law. Shops cannot accept foreign currency in return for goods or services.

Your notes

Currency Pairs

There is a convention to how we write exchange rates in a market.

👉 To write a currency pair, write both currency codes and separate them with a backward slash (/)

USD/INR



👉 The first currency is the **base currency**.

👉 The second currency is the **quote currency**.

👉 The base currency is given a value of 1.

👉 It has no fixed value.

👉 This example refers to 1 **USD**. We often refer to the base currency as the **reference element**.

Your notes

Currency Pairs

How do we *read* exchange rate quotes?

$$\text{USD/INR} = 80.00$$

☞ An exchange rate quote states the following:
how much you pay of the quote currency to buy 1 unit of the base currency.

☞ The quote above reads “Sell 80 INR to buy 1 USD”.

How do we denote the exchange rate if we are selling USD and buying INR?

Many of you will think that this will be the answer:

$$\text{INR/USD} = 0.0125$$

Your notes

Currency Rankings

You would be **WRONG**, *mostly*. This is because of the idea of **currency rankings**.

☞ Currency pairs are written regardless of whether the base currency is being bought or sold.

☞ There is a convention to write a currency pair, based off the rankings of currencies:

1. Euro, EUR (€)
2. Great British Pound Sterling, GBP (£)
3. Australian Dollar, AUD (\$ or A\$)
4. United States Dollar, USD (\$)
5. Canadian Dollar, CAD (\$ or C\$)
6. Swiss Franc, CHF (CHF)
7. Japanese Yen, JPY (¥)

LEFT: Rankings of Majors

☞ The 7 most important currencies are called ***the majors***.

☞ They are written first in a currency pair.

Your notes

Currency Rankings

Let's look at some examples of currency pair notation in action.

EUR/USD

GBP/JPY

EUR/AUD

CAD/JPY

AUD/USD

CAD/CHF

GBP/USD

EUR/CHF

1. Euro, EUR (€)
2. Great British Pound Sterling, GBP (£)
3. Australian Dollar, AUD (\$ or A\$)
4. United States Dollar, USD (\$)
5. Canadian Dollar, CAD (\$ or C\$)
6. Swiss Franc, CHF (CHF)
7. Japanese Yen, JPY (¥)

👉 **NO** correlation between *ranking* and *traded volume*.

👉 USD is still the most traded currency in the world.

👉 The list isn't fixed, so some deviations may be seen.

Your notes

Exotic currencies

But what about other currencies that aren't majors?

☞ Currencies that aren't majors are called **exotics**.

Some examples include the Chinese Yuan (**CHY**) and the Russian Ruble (**RUB**).

How do we trade exotics? How do we write them down in a currency pair?

☞ We pair up an exotic with a major currency!

CAD/CNY JPY/RUB

☞ Any currency pair that doesn't include the USD is known as a **cross-currency pair**.

☞ Any currency pair that includes the EUR is known as a **euro cross**.

GBP/JPY EUR/CHF

EUR/GBP EUR/CAD

Your notes

Selling and buying a currency pair

Let's return to selling **USD** and buying 1 **INR**.

$$\text{USD/INR} = 80.00$$

☞ The quote states that to buy 1 USD, you must pay 80 INR – you are **buying** the currency pair.

☞ It also means that to **buy** 80 INR, you must sell 1 USD – you are **selling** the currency pair.

☞ You can buy or sell a currency pair by only needing to specify the currency pair itself.

Selling the currency pair

Sell the base currency **USD**,
Buy the quote currency **INR**.

$$\text{USD/INR} = 80.00$$

Buy the base currency **USD**,
Sell the quote currency **INR**.

Buying the currency pair

Trading between exotic currencies

How do we trade between e.g. **CHY** and **RUB**?

👉 1. Trade between **CHY** and **RUB** directly.

👉 There's no conventional currency pair notation, so CHY/RUB or RUB/CHY are both valid.

👉 A lack of buyers and sellers may mean poor prices and a large bid-ask spread (i.e. lack of **liquidity**)

Chinese Yuan (CHY)

Russian Ruble (RUB)

👉 2. Buy an intermediary currency pair containing the exotic e.g. the cross-pair USD/CHY.

👉 Then, sell the USD/RUB to complete the CHY/RUB trade. Fees and commissions may be an issue.

Chinese Yuan (CHY)

United States Dollar (USD)

Russian Ruble (RUB)

Types of exchange markets

What types of markets exist?

☞ On the foreign exchange market, **currencies** are traded.

☞ Currencies are traded to be able to trade goods & services internationally and to invest abroad.

☞ Currencies are also traded by **speculation** – the belief of the price of a currency changing later.

☞ There are two types of exchange markets that trade currencies.

Spot Exchange Markets

Futures Exchange Markets

Your notes

Spot exchange markets

Here, we're talking about spot *foreign* exchange markets, but applies to all markets.

☞ Currencies are *traded right now* (i.e. 'on the spot')

A shop is a spot market for goods.

☞ You take '*immediate delivery of the asset*' (asset in FX market = currency being bought)

☞ An example of a spot *foreign* exchange market for retail investors is a Bureau de Change.



Alwyn Ladell, Flickr

☞ Some cases may require future delivery of the asset.

What type of market do you think facilitates this trade?

Your notes

Future exchange markets

Here, we're talking about future *foreign* exchange markets, but applies to all markets.

We will be talking about forwards and futures contracts in Lecture 4.

☞ Currencies are bought today but delivered at a time far *forward* in the future.

☞ This is facilitated by purchases of a **futures contract**.

☞ If the USD/INR falls to 70.00 in 6 months, Apple could sell the USD/INR today at 80.00.

☞ Apple will take delivery of their INR 6 months later at today's price regardless of any price changes.

Your notes

Exchange rate systems

Exchange rates are dynamic – they're not (normally) fixed at a value.

☞ For all trades, a supply and demand curve can be drawn.

☞ The equilibrium point is known as the exchange rate for two currencies.

☞ There are 3 general types of exchange rate systems:

Floating Exchange
Rate systems

*Price is determined by supply
and demand.*

Fixed Exchange
Rate systems

*Prices are fixed to a reference
and actively managed.*

Managed Exchange
Rate Systems

*Prices are both fixed to a
reference and left to float.*

Your notes

Floating exchange rate systems

You might recognise the concepts below already from earlier, which is good.

☞ If a currency is priced cheap, it will (generally) have a lot of demand from buyers.

☞ If a currency is priced high, it will (generally) have little demand from buyers.

☞ You can model this (generally linear) relationship using a demand curve (here for **EUR/GBP**):



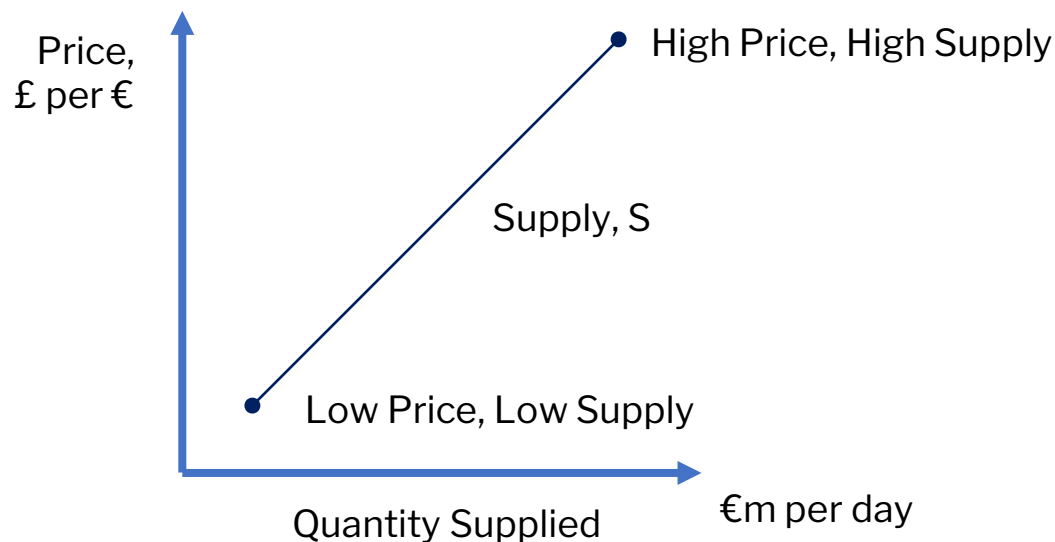
Floating exchange rate systems

What about the sellers, who are *supplying* the € to the market?

☞ If a currency is priced cheap, few suppliers will (generally) be prepared to supply the market with €.

☞ If a currency is priced high, many suppliers will (generally) be prepared to supply the market with €.

☞ You can model this (generally linear) relationship as well using a supply curve:



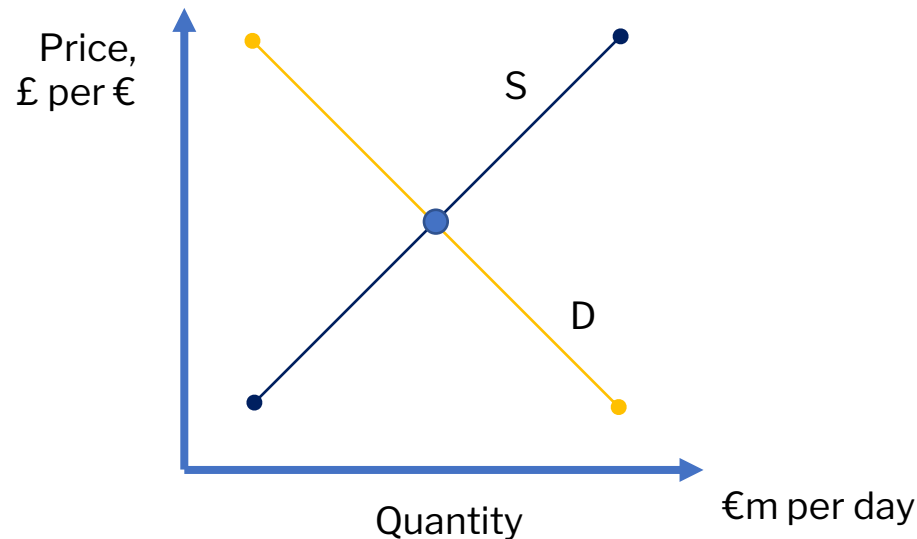
Floating exchange rate systems

What if we combine both the supply and demand curves together?

👉 By combining both S and D curves, you get a supply-demand curve.

👉 This curve is for the **EUR/GBP** currency pair.

👉 The intersection is the **equilibrium point** i.e. the current exchange rate.



Floating exchange rate systems

Why is a floating exchange rate system sometimes not deemed feasible?

☞ Central banks and government fiscal policy don't favour floating exchange rate systems.

☞ Floating exchange rates are determined by supply and demand.

GBP/USD

16 AUGUST 2021
1.3840

16 AUGUST 2022
1.2081

1 Y CHANGE, %
(12.71)

☞ In the example above, the pound has become '*weaker against the dollar*'.

☞ Exports are cheaper for **foreign** investors, but imports are more expensive for **domestic** investors.

Your notes

Purchasing Power Parity (PPP)

This is non-examinable, but useful to know.

☞ A unit of currency is any currency that there is 1 of e.g. 1 EUR, 1 GBP, 1 USD, 1 CHF, 1 JPY etc.

Consider for now that FX rates are *purely* determined by supply and demand.

GBP/USD

16 AUGUST 2022

1.2081

☞ Exchange rates can also be compared between two domestic currencies by a basket of goods.



Bread,
£1.20



Cereal,
£2.00



Milk,
£1.45

UK basket **TOTAL** £4.65

UK
Vs
US



Bread,
\$1.65



Cereal,
\$3.35



Milk,
\$2.05

US basket **TOTAL** \$7.05

Purchasing Power Parity (PPP)

Two different exchange rates at two different values. Why?

Why is it that two exchange rates exist here?

GBP/USD

16 AUGUST 2022

1.2081



Bread,
£1.20



Milk,
£1.45



Cereal,
£2.00

UK basket **TOTAL** £4.65

**GBP/
USD**
~ 1.50



Bread,
\$1.65



Milk,
\$2.05



Cereal,
\$3.35

US basket **TOTAL** \$7.05

☞ Exchange market rates are known as *nominal* exchange rates i.e. unrealistic and not a true rate.

☞ A basket of goods for two domestic currencies gives you a *real* exchange rate.

Purchasing Power Parity (PPP)

This all leads to one important concept: parity of purchasing power.

☞ The nominal and real exchange rates **should** be equal for equal purchasing power in the UK and US.

☞ This is known as the *Purchasing Power Parity (PPP) principle* → The **'law of one price'**.

☞ The **law of one price**: "Exchange rates should be set so an item costs one price everywhere"

☞ A classic PPP analysis is the *Big Mac PPP*.

☞ The Big Mac PPP also has a Big Mac *index*.



UK Price
£3.49

Switzerland Price
6.50 CHF

'Implied' (i.e. real) GBP/CHF exchange rate: 1.8624

Nominal GBP/CHF exchange rate (16/08/22): 1.1500
→ The CHF is overvalued with respect to GBP.

Fixed exchange rate systems

Why do some central banks insist on a fixed exchange rate system?

☞ They can provide a clear and well-defined exchange rate to foreign and domestic investors.

☞ A management of the money supply helps maintain **low inflation** and **fiscal expansion**.

☞ Currencies used to be fixed to an exchange rate with gold → the ‘gold standard’

☞ Some countries use a fixed exchange rate system:

☞ These currencies are ‘fixed’ or ‘pegged’ to the USD!



**Saudi Arabia
(SAR)**



**Hong Kong
(HKD)**



**United Arab
Emirates (DIR)**

Your notes

Gold standard

Where did fixed exchange rate systems start?

👉 A domestic currency used to be fixed in proportion to the central bank's store of gold.

1870s – 1914, 1920s – 1932

15 grams of gold = £2.06 or \$10.00

GBP/USD

≈ 4.87

This was the **gold standard**.

(Also available in France, Germany etc.)

👉 The money supply was also *fixed*, so fiscal expansion (printing more money) was not possible.

👉 **Bretton Woods, 1944** – establishment of the fixed standard to the US dollar.*

👉 Countries fixed their currency to the USD, which in turn was fixed to the Federal Gold Reserve.

👉 Abandoned in 1973 by Richard Nixon due to inflationary policy and dwindling gold reserves.
The US dollar was no longer 'backed' by gold (i.e. no USD → gold exchange) – led to a 'fiat' currency.

*Bretton Woods isn't actually a fixed exchange system – this is a gross simplification.

Fiat currency

Most currencies in the world are known as ‘fiats’ – what do we mean by a **fiat** currency?

☞ Previously, currencies could be guaranteed an exchange rate with gold.

☞ Since the currency ⇌ gold exchange is no longer valid, the USD is not backed by anything.

☞ A currency *not* backed by anything is a **fiat** currency. (*Fiat* – an order given by someone in power.)

☞ Nixon declared an order that the USD has value as it is a valid method of purchase in the USA.

☞ The USD is not ‘backed’ by any real commodity e.g. as gold, so it has a believed, ‘hypothetical’ value.

Your notes

Managed Exchange Rate Systems

What if we try a *hybrid* between a floating and a fixed exchange rate system?

👉 **Managed Exchange Rate Systems** combine supply-demand forces and government intervention.

👉 There are three types of managed float systems:

Adjustable peg systems

Crawling peg systems

Managed float systems

👉 They are also called **Hybrid Exchange Rate Systems**.

👉 Be aware that managed exchange rate systems **ARE NOT** fixed exchange rate systems!

Your notes

Managed Exchange Rate Systems

Adjustable peg systems are the first types of managed exchange rate systems you will see.

👉 When a government fixes its domestic currency against another currency, it is known as a **peg**.

👉 In the short-term, the peg is normally fixed.

👉 In the long-term, pegs are normally *adjusted*.

👉 One common example of a managed exchange rate system is the *Bretton Woods* system.

👉 The exchange rate is kept in a specific band of limits on a long-term basis (as you'll see next).

👉 This is achieved through manipulation of supply and demand in the markets.

Your notes

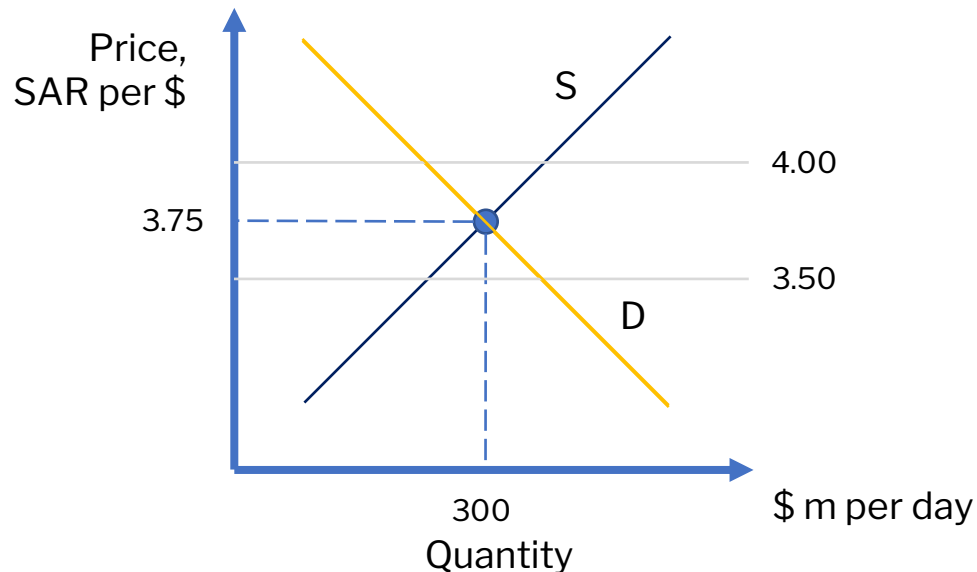
Managed Exchange Rate Systems

How can we model adjustable peg systems?

👉 Consider the basic supply-demand graph for USD/SAR, and add some limits to the exchange rate.

👉 USD/SAR trades at 3.75 SAR, with \$300m supplied to the market every day.

What happens if the demand for the USD increases?



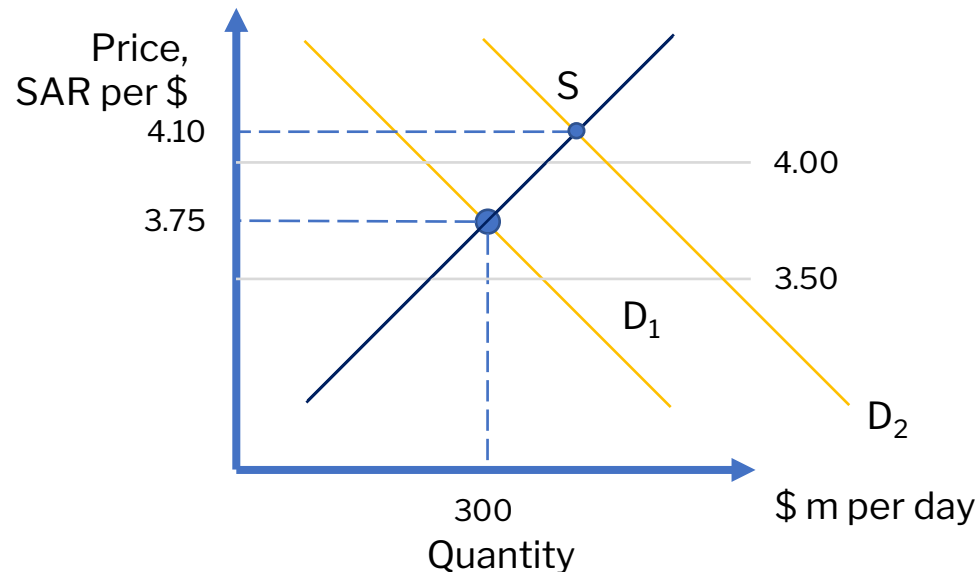
Managed Exchange Rate Systems

Model the demand curve shifting up.

☞ The demand curve shifts right, labelled **D₂**

☞ The USD has now got stronger at 4.10 SAR, above the 4.00 SAR limit.

How do we recover the price back to the pre-defined limits?

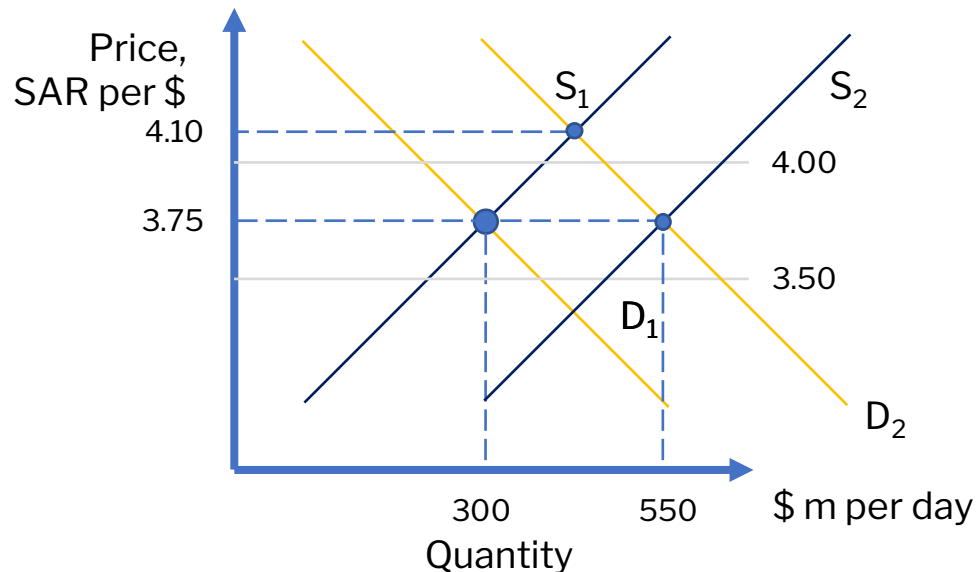


Managed Exchange Rate Systems

Shift the supply curve to the right.

👉 The Saudi central bank will sell its own USD reserves to flood the market, increasing supply.

👉 A new curve S_2 is established, shifted right to bring the price back to pre-defined limits.



Managed Exchange Rate Systems

Our tour of FX ends with an overview of crawling pegs and managed float systems.

➡ A **crawling peg system** is an extension of the adjustable peg system.

➡ The limits are allowed to move ('crawl') to reflect recent economic and market momentum.

➡ This is achieved by the Chinese government with the Chinese Yuan.

➡ If the price of the currency slowly is increasing, the limits are slowly moved up in turn.

➡ The peg is still in-place to restore prices (as per the adjustable peg system).

Your notes

Managed Exchange Rate Systems

And now the finale: managed float systems.

👉 A **managed float system** allows currency prices to be determined by supply and demand.

👉 The local currency is **NOT** pegged to a foreign currency by the central bank.

👉 Government intervention is **ALLOWED** to restore the price to a desired value.

👉 Price restoration is achieved by altering supply and demand through buying and selling.

👉 This is the most common exchange rate system e.g. GBP, EUR, JPY etc.

Your notes